Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The reaction $2\ \mathrm{N_2O_5} \rightleftharpoons 2\ \mathrm{N_2O_4} + \mathrm{O_2}$ is

(A) bimolecular and of second order

(B) unimolecular and of first order

(C) bimolecular and of first order

(D) bimolecular and of zero order

Q.2

For the reaction system :
$2\mathrm{NO(g)} + \mathrm{O_2(\ g)} \rightarrow 2\mathrm{NO_2(\ g)}$ volume suddenly reduced to half its value by increasing the pressure on it. If the reaction is of first order with respect to $\mathrm{O_2}$ and second order with respect to NO, the rate of reaction will

(A) diminish to one-eighth of its initial value

(B) increase to eight times of its initial value

(C) increase to four times of its initial value

(D) diminish to one-fourth of its initial value

Q.3

Units of rate constant of first and zero order reactions in terms of molarity M unit are respectively

(A) $\mathrm{sec^{-1}, M \cdot sec^{-1}}$

(B) $\mathrm{sec^{-1}, M}$

(C) $\mathrm{M \cdot sec^{-1}, sec^{-1}}$

(D) $\mathrm{M, sec^{-1}}$

Q.4

A reaction rate constant is given by
$k = 1.2 \times 10^{14} e^{-25000/\mathrm{RT}} \mathrm{sec^{-1}}$. It means

(A) $\log k$ versus $\log \mathrm{T}$ will give a straight line with a slope as -25000

(B) $\log k$ versus T will give a straight line with slope as 25000

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(C) $\log k$ versus $1/\mathrm{T}$ will give a straight line with slope as $-25000/\mathrm{R}$

(D) None of these

Q.5

The time required for 10% completion of a first order reaction at $298\ \mathrm{K}$ is equal to that required for its 25% completion at $308\ \mathrm{K}$. If the preexponential factor for the reaction is $3.56 \times 10^{9}\ \mathrm{s}^{-1}$, the activation energy at $318\ \mathrm{K}$ is:

(A) $18.39\mathrm{kcalmol}^{-1}$

(B) $20\mathrm{kcalmol}^{-1}$

(C) $16\mathrm{kcalmol}^{-1}$

(D) $21.5\mathrm{kcalmol}^{-1}$

Q.6

In a reaction $A \rightarrow$ Products, when start is made from $8.0 \times 10^{-2}\mathrm{M}$ of A, half-life is found to be 120 minute. For the initial concentration $4.0 \times 10^{-2}\mathrm{M}$, the half-life of the reaction becomes 240 minute. The order of the reaction is:

(A) zero

(B) one

(C) two

(D) 0.5

Q.7

Select the rate law that corresponds to data shown for the following reaction $\mathrm{A} + \mathrm{B} \longrightarrow$ products.

Exp.	[A]	[B]	Initial rate
1	0.012	0.035	0.1
2	0.024	0.070	0.8
3	0.024	0.035	0.1
4	0.012	0.070	0.8

(A) rate $= \mathrm{k}[\mathrm{B}]^{3}$

(B) rate $= \mathrm{k}[\mathrm{B}]^{4}$

(C) rate $= \mathrm{k}[\mathrm{A}][\mathrm{B}]^{3}$

(D) rate $= \mathrm{k}[\mathrm{A}]^{2}[\ \mathrm{B}]^{2}$

Q.8

The initial rates of reaction

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$3\ \mathrm{A} + 2\ \mathrm{B} + \mathrm{C} \longrightarrow$ Products, at different initial concentrations are given below:

Initial rate, $\mathbf{Ms}^{-1}$	$[\mathbf{A}]_0, \mathbf{M}$	$[\mathbf{B}]_0, \mathbf{M}$	$[\mathbf{C}]_0, \mathbf{M}$
5.0×10^{-3}	0.010	0.005	0.010
5.0×10^{-3}	0.010	0.005	0.015
1.0×10^{-2}	0.010	0.010	0.010
1.25×10^{-3}	0.005	0.005	0.010

The order with respect to the reactants, A, B and C are respectively

(A) $3, 2, 0$

(B) $3, 2, 1$

(C) $2, 2, 0$

(D) $2, 1, 0$

Q.9

Consider a reaction $aG + bH \rightarrow$ Products. When concentration of both the reactants G and H is doubled, the rate increases by eight times. However, when concentration of G is doubled keeping the concentration of H fixed, the rate is doubled. The overall order of the reaction is

(A) 0

(B) 1

(C) 2

(D) 3

Q.10

For the reaction $\mathrm{A} + \mathrm{B} \longrightarrow$ products, it is observed that:
(1) On doubling the initial concentration of A only, the rate of reaction is also doubled and
(2) On doubling the initial concentrations of both A and B, there is a change by a factor of 8 in the rate of the reaction. The rate of this reaction is given by:

(A) rate $= \mathrm{k}[\mathrm{A}][\mathrm{B}]^2$

(B) rate $= \mathrm{k}[\mathrm{A}]^2[\ \mathrm{B}]^2$

(C) rate $= \mathrm{k}[\mathrm{A}][\mathrm{B}]$

(D) rate $= \mathrm{k}[\mathrm{A}]^2[\ \mathrm{B}]$

Q.11

The order of a reaction, with respect to one of the reacting component Y, is zero. It implies that:

(A) the reaction is going on at a constant rate

(B) the rate of reaction does not vary with temperature

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(C) the reaction rate is independent of the concentration of Y

(D) the rate of formation of the activated complex is zero

Q.12

The rate law for the reaction below is given by the expression $\mathrm{k}[\mathrm{A}][\mathrm{B}]$

$\mathrm{A}+\mathrm{B} \rightarrow$ Product

If the concentration of B is increased from 0.1 to 0.3 mole, keeping the value of A at 0.1 mole, the rate constant will be:

(A) $3\mathrm{k}$

(B) $9\mathrm{k}$

(C) $\mathrm{k}/3$

(D) k

Q.13

The half-life of a reaction is inversely proportional to the square of the initial concentration of the reactant. Then the order of the reaction is

(A) 0

(B) 1

(C) 2

(D) 3

Q.14

For a reaction $\mathrm{A}+2\ \mathrm{B} \rightarrow \mathrm{C}$, the amount of formed by starting the reaction with 5 moles of A and 8 moles of B is

(A) 5 moles

(B) 8 moles

(C) 16 moles

(D) 4 moles

Q.15

Consider the two hypothetical reactions given below:

I $aA \rightarrow$ Products, $\quad k = x\ \mathrm{mol}^{-1}\ \mathrm{L}\ \mathrm{min}^{-1}$

II $bB \rightarrow$ Products, $\quad k = y\ \mathrm{min}^{-1}$

The half-lives of both the reactions are the same, equal to $1\mathrm{hr}$ when molar concentration of the reactant is

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$1.0\mathrm{M}$ in each case. If these reactions are started at the same time taking $1\mathrm{M}$ of the reactant in each case, the ratio $[A]/[B]$ after $3\mathrm{hr}$ will be :

(A) 0.5

(B) 4

(C) 1

(D) 2

Q.16

The rate constant of a zero order reaction is $2.0 \times 10^{-2}\ \mathrm{mol\ L^{-1}\ s^{-1}}$. If the concentration of the reactant after 25 seconds is $0.5\mathrm{M}$. What is the initial concentration?

(A) $0.5\mathrm{M}$

(B) $1.25\mathrm{M}$

(C) $12.5\mathrm{M}$

(D) $1.0\mathrm{M}$

Q.17

The rate of a reaction increases four-fold when the concentration of reactant is increased 16 times. If the rate of reaction is $4 \times 10^{-6}\ \mathrm{mol\ L^{-1}\ s^{-1}}$ when the concentration of the reactant is $4 \times 10^{-4}\ \mathrm{mol\ L^{-1}}$. The rate constant of the reaction will be

(A) $2 \times 10^{-4}\ \mathrm{mol^{1/2}\ L^{-1/2}\ s^{-1}}$

(B) $1 \times 10^{-2}\ \mathrm{s^{-1}}$

(C) $2 \times 10^{-4}\ \mathrm{mol^{-1/2}\ L^{1/2}\ s^{-1}}$

(D) $25\ \mathrm{mol^{-1}\ L\ min^{-1}}$

Q.18

The reaction of A_2 and B_2 follows the equation

$$\mathrm{A_2(\ g) + B_2(\ g) \rightarrow 2AB(g)}$$

The following data were observed

$$\begin{array}{|c|c|c|}
\hline \left[\mathrm{A}_{2}\right]_{0} & \left[\mathrm{~B}_{2}\right]_{0} & \text{Initial rate of appearance of } \mathrm{AB}(\mathrm{g})\left(\mathrm{in}\ \mathrm{Ms}^{-1}\right) \\
\hline 0.10 & 0.10 & 2.5 \times 10^{-4} \\
\hline 0.20 & 0.10 & 5 \times 10^{-4} \\
\hline 0.20 & 0.20 & 10 \times 10^{-4} \\
\hline
\end{array}$$

The value of rate constant for the above reaction is:

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(A) 2.5×10^{-4}

(B) 2.5×10^{-2}

(C) 1.25×10^{-2}

(D) None of these

Q.19

A chemical reaction was carried out at $300\ \mathrm{K}$ and $280\ \mathrm{K}$. The rate constants were found to be k_1 and k_2 respectively. then

(A) $k_2 = 4k_1$

(B) $k_2 = 2k_1$

(C) $k_2 = 0.25k_1$

(D) $k_2 = 0.5k_1$

Q.20

The reaction $\mathrm{A(g) + 2\ B(\ g) \rightarrow C(g) + D(g)}$ is an elementary process. In an experiment, the initial partial pressure of A and B are $\mathrm{P_A} = 0.60$ and $\mathrm{P_B} = 0.80\ \mathrm{atm}$. When $\mathrm{P_C} = 0.2\ \mathrm{atm}$ the rate of reaction relative to the initial rate is

(A) $1/48$

(B) $1/24$

(C) $9/16$

(D) $1/6$

Answers & Solutions

Q.1 Answer:

bimolecular and of first order

Solution:

It is bimolecular first order reaction since Rate $\propto [\mathrm{N_2O_5}]$

Q.2 Answer:

increase to eight times of its initial value

Solution:

$r = k\,[\mathrm{O_2}]\,[\mathrm{NO}]^2$. **When the volume is reduced to** $1/2$**, the conc. will double**

$\therefore$ **Newrate** $= k\,[2\mathrm{O_2}]\,[2\mathrm{NO}]^2 = 8k\,[\mathrm{O_2}]\,[\mathrm{NO}]^2$

The new rate increases to eight times of its initial.

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Q.3 Answer:

$\sec^{-1}, \mathrm{M} \cdot \sec^{-1}$

Solution:

For a zero order reaction. rate $= k[A]^{\circ}$ **i.e., rate** $= k$ **hence unit of** $k = \mathrm{M} \cdot \sec^{-}$

For a first order reaction. rate $= k[A]$ $k = \mathrm{M} \cdot \sec^{-1} / \mathrm{M} = \sec^{-1}$

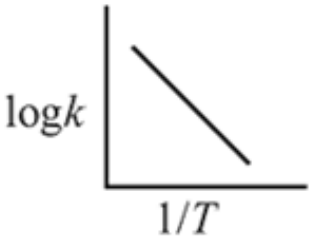
Q.4 Answer:

$\log k$ versus $1/\mathrm{T}$ will give a straight line with slope as $-25000/\mathrm{R}$

Solution:

$k = 1.2 \times 10^{14} e^{-25000/\mathrm{RT}} \sec^{-1}$ or

$\log k = \log 1.2 \times 10^{14} - \frac{25000}{\mathrm{R}} \cdot \frac{1}{\mathrm{T}}$



Equation of straight line

slope $= -\frac{25000}{R}$

Q.5 Answer:

$18.39 \mathrm{kcalmol}^{-1}$

Solution:

Let the initial concentration $(A) = 100$

Final concentration at $298\ \mathrm{K} = 100 - 10 = 90$

Final concentration at $308\ \mathrm{K} = 100 - 25 = 75$

Substituting the values in the 1 st order rate reaction

$t = \frac{2.303}{k_{298}} \log \frac{100}{90}$ **...(i)**

$t = \frac{2.303}{k_{308}} \log \frac{100}{75}$ **...(ii)**

From (i) and (ii) $\frac{k_{308}}{k_{208}} = 2.73$

Substituting the value in the following relation

$E_a = \frac{2.303 R \times T_1 \times T_2}{T_2 - T_1} \log \frac{k_2}{k_1}$

$= \frac{2.303 \times 8.314 \times 298 \times 308}{308 - 298} \log 2.73$

$E_o = 76.62\ \mathrm{kJ\ mol}^{-1} = 18.39 \mathrm{kcalmol}^{-1}$

Q.6 Answer:

two

Solution:

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$\frac{\left(t_{1/2}\right)_1}{\left(t_{1/2}\right)_2}=\left(\frac{a_2}{a_1}\right)^{n-1} ; \frac{120}{240}=\left(\frac{4\times10^{-2}}{8\times10^{-2}}\right)^{n-1} ; n=2$

Q.7 Answer:

rate $=\mathrm{k}[\mathrm{B}]^3$

Solution:

From data 1 and 3, it is clear that keeping (B) const, When [A] is doubled, rate remains unaffected. Hence rate is independent of $[\mathrm{A}]$. from 1 and 4, keeping $[\mathrm{A}]$ constant, when [B] is doubled, rate become 8 times.

Hence rate $\propto[\mathrm{B}]^3$.

Q.8 Answer:

$2,1,0$

Solution:

From 1st and 2nd sets of data - no change in rate is observed with the change in concentration of ' C'. So the order with respect to ' C ' is zero.

From 1st and 4th sets of data

Dividing eq. (4) by eq. (1)

$\frac{1.25\times10^{-3}}{5.0\times10^{-3}}=\left[\frac{0.005}{0.010}\right]^x$

or $0.25=(0.5)^x$ **or** $(0.5)^2=(0.5)^x$

$\therefore \mathrm{x}=2$

The order with respect to ' A ' is 2 from the 1st and 3rd sets of data dividing eq. (1) by eq. (3)

$\frac{5.0\times10^{-3}}{1.0\times10^{-2}}=\left[\frac{0.005}{0.010}\right]^y$

or $(0.5)^1=(0.5)^y \Rightarrow y=1$

The order with respect to ' B ' is 1

So the order with respective the reactants A, B and C is 2,1 and 0.

Q.9 Answer:

3

Solution:

Overall order = sum of orders w.r.t each reactant.

Let the order be x and y for G and H respectively

\begin{array}{|c|c|c|c|}

**\hline Exp.No. & [G]mole litre ^{-1} & [H]mole litre ^{-1} & rate(mole litre ^{-} time ^{-1}) **

**\hline 1 & a & b & r **

**2 & 2 a & 2 b & 8 r **

**3 & 2 a & b & 2 r **

\hline

\end{array}

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∵ For (1) and (3), the rate is doubled when conc. of G is doubled keeping that of H constant i.e., rate $\propto [\mathrm{G}]$ $\therefore x = 1$
From (2) and (3), $\mathrm{y} = 2$
$\therefore$ Overall order is 3.

Q.10 Answer:

rate $= \mathrm{k}[\mathrm{A}][\mathrm{B}]^2$

Solution:

When concentration of A is doubled, rate is doubled. Hence order with respect to A is one. When concentrations of both A and B are doubled, rate increases by 8 times hence order with respect to B is 2 .
$\therefore$ rate $= \mathrm{k}[\mathrm{A}]^1[\mathrm{B}]^2$

Q.11 Answer:

the reaction rate is independent of the concentration of Y

Solution:

Let us consider a reaction, $xX + yY \longrightarrow aA + bB$
rate $= [X]^x[Y]^y$
It is given that order of reaction w.r.t. component Y is zero.
Hence, rate $= [X]^x$
i.e., rate becomes independent of the concentration of Y.

Q.12 Answer:

k

Solution:

Rate constant is independent of concentration.

Q.13 Answer:

3

Solution:

$t_{1/2} \propto \frac{1}{a^2}$
We know that $t_{1/2} \propto \frac{1}{a^{n-1}}$
i.e. $n = 3$ Thus reaction is of 3rd order

Q.14 Answer:

4 moles

Solution:

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$\mathrm{A} + 2\,\mathrm{B} \longrightarrow \mathrm{C}$

1 mole of A reacts with 2 moles of B to give $1\ \mathrm{mol}$ of C.

$\therefore$ 5 moles of A would react with 10 moles of B to give 5 moles of C. But, only 8 moles of B are available

$\therefore$ B acts as a limiting reagent.

2 moles of B gives 1 mole of C

$\therefore$ 8 moles of B will give $1/2 \times 8 = 4$ moles of C.

Q.15 Answer:

2

Solution:

Units of k indicate that reaction I is of second order and reaction II is first order. For I reaction,

$t_{1/2} \propto 1/a$**, first** $t_{1/2=}1\mathrm{hr}$**, second** $t_{1/2} = 2\mathrm{hr}$

$[A] = 1\mathrm{M} \xrightarrow{1\mathrm{hr}} 0.5\mathrm{M} \xrightarrow{2\mathrm{hr}} 0.25\mathrm{M}$

$[B] = 1\mathrm{M} \xrightarrow{1\mathrm{hr}} 0.5\mathrm{M} \xrightarrow{1\mathrm{hr}}$

$0.25\mathrm{M} \xrightarrow{1\mathrm{hr}} 0.125\mathrm{M}$

$\frac{[A]}{[B]} = \frac{0.25\mathrm{M}}{0.125\mathrm{M}} = 2$

Q.16 Answer:

$1.0\mathrm{M}$

Solution:

For a zero order reaction

Rate constant $= \mathrm{k} = \frac{a-x}{t}$

$2 \times 10^{-2} = \frac{a-0.5}{25}$

$a - 0.5 = 0.5$

$a = 1.0\mathrm{M}$

Q.17 Answer:

$2 \times 10^{-4}\ \mathrm{mol}^{1/2}\ \mathrm{L}^{-1/2}\ \mathrm{s}^{-1}$

Solution:

Rate $\propto \sqrt{\text{ Concentration }} = \mathrm{k}\sqrt{\text{ Concentration }}$

$\mathrm{k} = \frac{\text{Rate}}{(\text{ Concentration })^{1/2}}$

$= \frac{4\times10^{-6}}{(4\times10^{-4})^{1/2}} = \frac{4\times10^{-6}}{2\times10^{-2}}$

$= 2 \times 10^{-4}\ \mathrm{mol}^{1/2}\ \mathrm{L}^{-1/2}\ \mathrm{s}^{-1}$

Q.18 Answer:

1.25×10^{-2}

Solution:

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Order w.r.t. $\mathrm{A} = 1$; **order w.r.t.** $\mathrm{B} = 1$

Rate $= \frac{1}{2}\frac{\mathrm{d(AB)}}{\mathrm{dt}} = \mathrm{k_r[A][B]}$

$\frac{1}{2} \times \left(2.5 \times 10^4\right) = \mathrm{k_r}(0.1)(0.1)$

$\mathrm{k_r} = 1.25 \times 10^{-2}$

Q.19 Answer:

$k_2 = 0.25k_1$

Solution:

The rate constant doubles for $10^\circ\mathrm{C}$ **rise in temperature. For** $20^\circ\mathrm{C}$ **rise, the rate constant will be 4 times** $\therefore\ k_1 = 4k_2$ **or** $k_2 = 0.25k_1$

Q.20 Answer:

$1/6$

Solution:

$\mathrm{R_1 = K[A][B] = K[0.6][0.80]^2}$

After reaction

A	$+2\ \mathrm{B} \longrightarrow$	C	$+\mathrm{D}$
$0.6 - 0.2$	$0.8 - 0.4$	0.2	0.2
0.4	0.4	0.2	0.2

$\frac{R_2}{R_1} = \frac{K(0.4)(0.4)^2}{(0.6)(0.8)^2} = \frac{1}{6}$

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